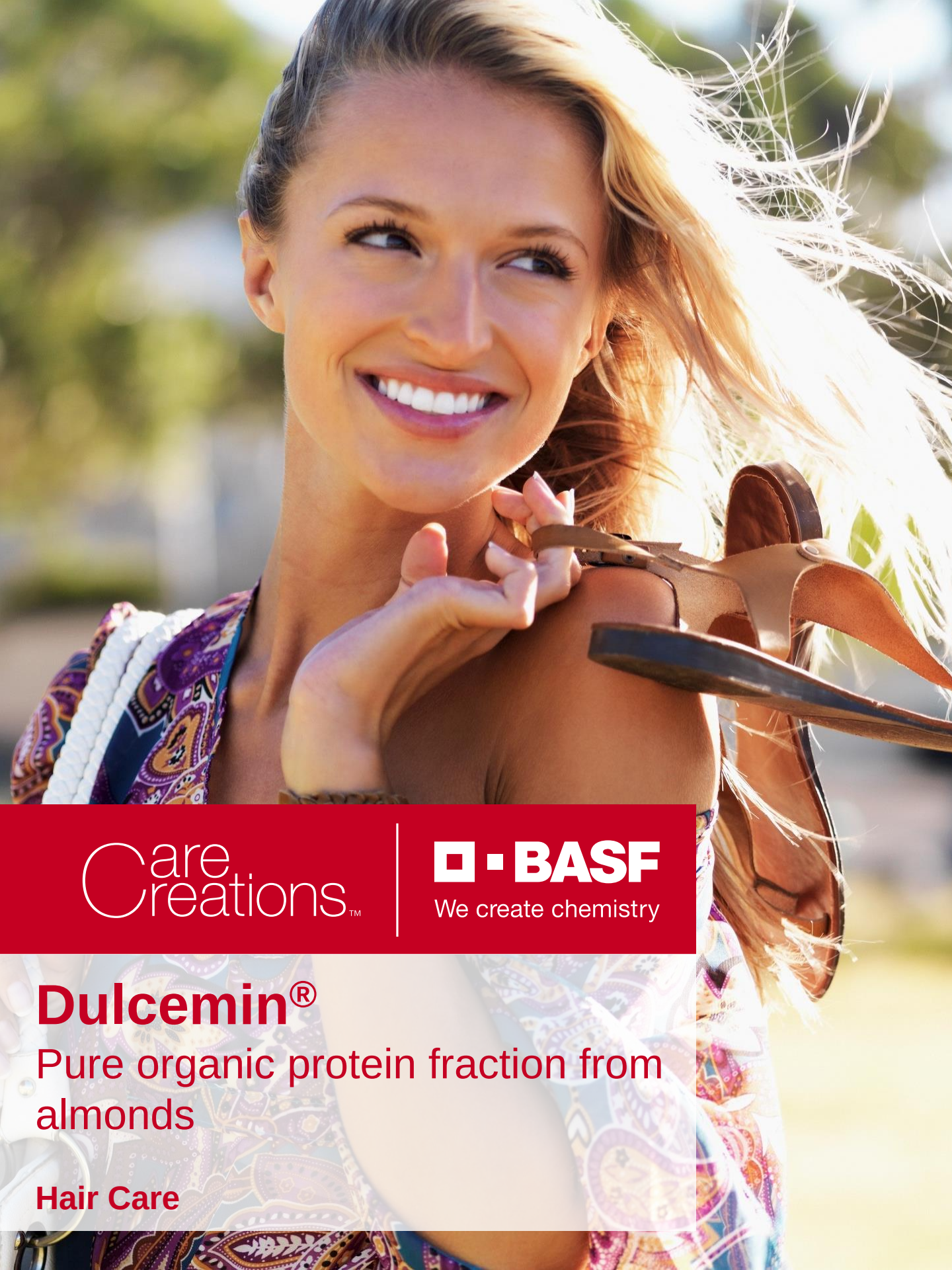


Care Creations™

BASF
We create chemistry

Dulcemin®

Pure organic protein fraction from almonds

Hair Care

Geographic origin and historic background

Cultivation of almond trees or *Prunus amygdalus* var. *dulcis* L., (*Rosaceae*) has increased considerably in recent years. Like the apricot, the peach and the plum trees, the almond tree originated in the Iran-Afghan region, from where they have spread around the Mediterranean. Today, California and Spain are the main producers.

The use of almond seeds for preparing drugs and cosmetics dates back to ancient times. In the 19th century, Osborne extracted a protein from almonds which he called Amandin. Sweet almond oil is a prized excipient. The seed was used to make emulsions, such as almond milk. This benefit derives from one of its protein constituents: emulsin. The almond seed is also a food with high nutritional value because of its lipid, protein, sugar and mineral content.

Protein fraction

The protein fraction of almond seed extract is most interesting for cosmetic care; it contains three constituent proteins:

- *Emulsin*, a complex of proteins with phosphatasic and β-glucosidasic enzymatic activity
- *Amandoalbumin* with MW ~150,000 Daltons
- *Amandoglobulin* with MW ~300,000 Daltons

Glutamic acid is the predominant amino acid in almond proteins. Sucrose is linked in a glycoprotein form. Many cosmetic products just contain sweet almond oil or simple hydrolysates of almond protein.

Definition / Composition

Dulcemin is a purified glycoprotein active ingredient, extracted from seeds of *Prunus amygdalus var. dulcis L*. Dulcemin contains a pure, stable amandoglobulin fraction. Dulcemin is innovative and exclusive, differing from common sweet almond actives such as partial hydrolysates of sweet almond proteins. Its main component, the glycoprotein amandoglobulin, consists of a native protein and an oligosaccharide (saccharose) conju-gated with the protein through a covalent bond. The average molecular weight of the proteins in Dulcemin, determined by HPLC, is ~290,000 Daltons. Thus Dulcemin may be considered a botanical analogue of collagen, another glycoprotein with a MW of ~300,000 Daltons.

Brief Overview: Dulcemin PW BIO LS 9903



INCI: Prunus Amygdalus Dulcis Protein

China registration: INCI name is listed in “Inventory of Existing Cosmetic Ingredient in China 2015” (IECIC 2015). CAS number is listed (or exempted) in “Inventory of Existing Chemical Substances in China 2013” (IECSC 2013).

Appearance: Beige powder with a weak characteristic odor.

Preservative: None

Recommended dosage: 0.1 – 0.4 % (15,4 times more concentrated than predecessor version Dulcemin LS 8594)

Mode of incorporation: Prepare a mother solution in water at 10 % in weight at 40°C (or room temperature by turbine stirring). Introduce this solution in the finishing process below 50°C or at room temperature for cold processing. Suitable pH 3 to 6.

Amino acid composition of the glycoprotein of Dulcemin

Amino Acid residue	Content (%)	Amino Acid residue	Content (%)
Aspartic acid	3.20	Tyrosine	2.55
Glutamic acid	41.92	Valine	3.80
Serine	3.95	Methionine	1.28
Glycine	3.85	Cysteine	1.53
Histidine	1.75	Isoleucine	0.58
Arginine	3.67	Leucine	6.65
Threonine	2.47	Phenylalanine	5.30
Alanine	2.10	Lysine	1.90
Proline	13.50		

Hair benefits

Dulcemin is suitable as a conditioning and nutritious agent for hair that is split, dull, brittle, thin or dry. Repairs hair weakened by bleaching, coloring or perming. Reduces irritation caused by some surfactants.

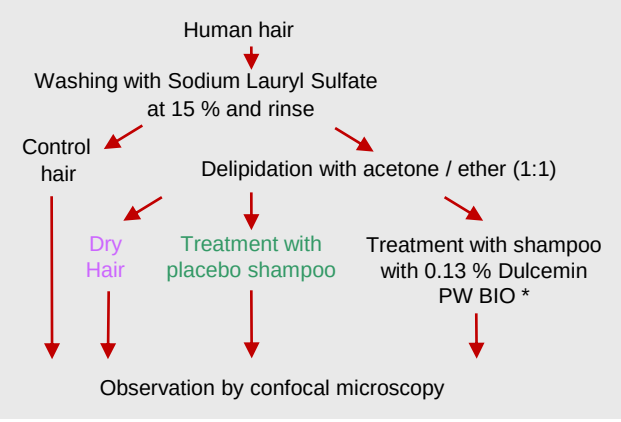
Efficacy

Film forming effect on human hair (confocal microscopy)

Aim

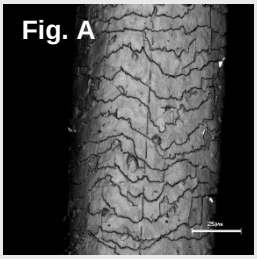
To demonstrate the conditioning and protecting effect of Dulcemin in a shampoo on dry human hair.

Protocol

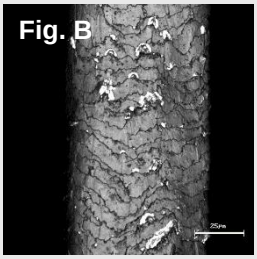


Results

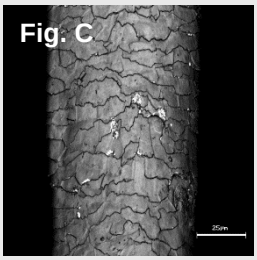
Confocal microscopic images



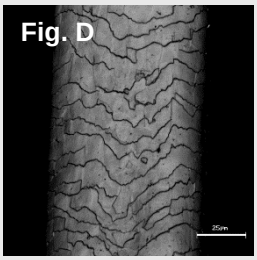
Control hair



Dry hair



Placebo shampoo



Shampoo containing 0.13 % Dulcemin PW BIO *

Results

Hair washed with a solution of sodium lauryl sulfate (SLS) has a healthy look with some cuticle scales being slightly uplifted. The lifted scales can be seen in confocal microscopy as white "streaks" on hair (Fig. A = control hair). Hair delipidated with acetone/ether serves as a model of dry hair. Drying hair causes cuticle damage in which ends of cuticle scales are uplifted (Fig. B). Hair treatment with the placebo shampoo has slightly improved the state of the hair surface (Fig. C). The improvement of surface texture is greater after treatment with Dulcemin. The scales are no longer lifted up, are very regular, and the hair has a smooth look as if it was coated with a protecting film (Fig. D).

* Test done with 2 % Dulcemin LS 8594 (aqueous solution, 15,4 x less concentrated than Dulcemin PW BIO LS 9903)

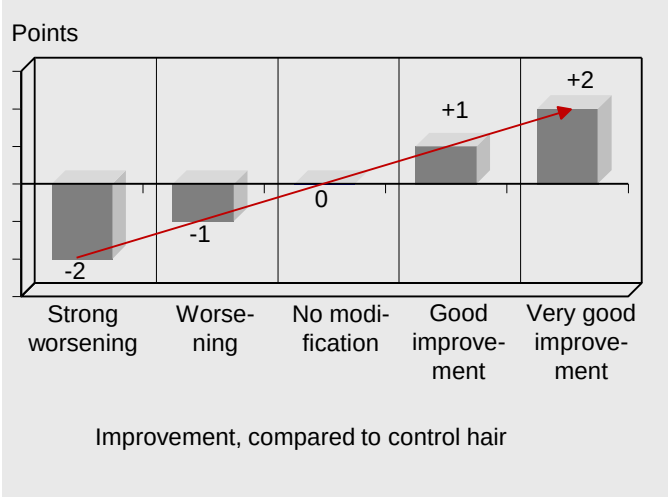
Improvement of sensorial properties of hair

Sensorial evaluation of hair benefits after treatment with a shampoo containing Dulcemin

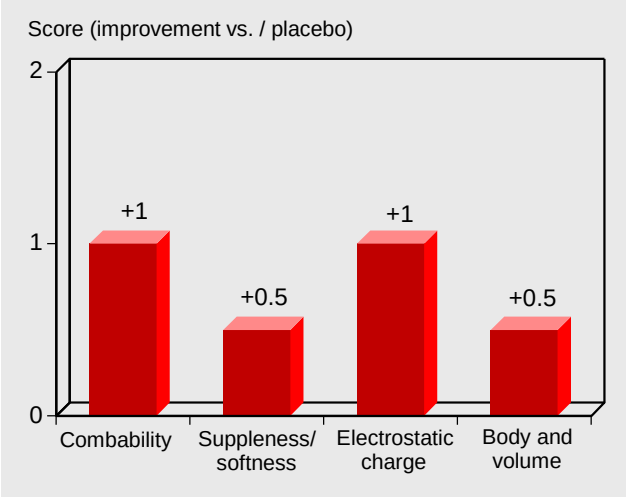
Protocol

Several sensorial parameters were evaluated on hair locks treated with placebo shampoo or a shampoo containing 0.13 % Dulcemin *.

Evaluation scale



Result



* Test done with 2 % Dulcemin LS 8594 (aqueous solution, 15,4 x less concentrated than Dulcemin PW BIO LS 9903)

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